TORCS AI Racing Competition

Customise Your TORCS

Personalise your car and circuit to stand out on race day

What you will achieve by following this guide:

✓ Change your car's paint scheme (livery)

✓ Add your own logo or number to your car

✓ Replace track banners with custom artwork

✓ Add multiple livery variants for different races

Who is this guide for?

Students who have already completed the TORCS Container Setup Guide and can run the simulator. You do not need any prior graphics or 3D experience — every step is explained from scratch.

0. Background — How TORCS Uses Textures

Everything you see in TORCS — cars, barriers, grandstands, banners — is a 3D model with a flat image (called a texture) wrapped around it. Swapping that image is all you need to do to change how something looks. No 3D modelling required.

Textures in TORCS are stored as .rgb files (SGI image format). GIMP, which is free and open-source, can open and save .rgb files directly. PNG files are also supported in most modern TORCS builds.

0.1 The golden rules of TORCS textures

- Dimensions must be powers of two — 256, 512, 1024, 2048. A 500 × 500 image will not load correctly.
- Keep the same filename as the original — TORCS loads textures by name, so renaming breaks the link.
- Always back up the original file before replacing it.
- After editing, restart TORCS completely — it caches textures at startup.

1. Changing Your Car Livery

A livery is the paint scheme applied to your car. TORCS stores livery textures in two different locations, and understanding which one to edit is the key to getting your changes to appear correctly.

1.1 Two locations, one texture — which to edit

You will find a livery texture file in two places inside the container. They look identical but serve different purposes:

Location	Path	Purpose
cars/ folder	`cars/car1-trb1/car1-trb1.rgb`	Master template. The factory paint job shipped with the car model. Used as a fallback only.

drivers/ folder	`drivers/src_server/0/car1-trb1.rgb`	Driver override. Loaded first during a race. **This is the one you should edit.**

Think of it like this: the cars/ texture is the factory paint job, and the drivers/ texture is a vinyl wrap applied on top. TORCS always checks for the wrap first. If it finds one, the factory paint is never seen. If no wrap exists, it falls back to the factory paint. Editing only the cars/ copy will have no visible effect during a race because your driver's own copy takes priority.

1.2 Copy the driver texture out of the container

Copy the texture from your driver's folder — this is the file that is actually loaded during a race:

```
podman cp
<container_name>:/usr/local/share/games/torcs/drivers/src_server/0/car1-trb1.rgb .
```

Tip: Replace <container_name> with the name of your running container. You can find it by running: podman ps

1.3 Edit the texture in GIMP

1. Open GIMP and go to File > Open. Select your .rgb file.
2. The image shows the flattened surface of the car body. Doors, bonnet, and roof are arranged as flat panels.
3. Paint over the panels with your chosen colours, add your team name, car number, or sponsor logos.
4. When finished, go to File > Export As. Keep the filename exactly the same and choose SGI Image (*.rgb) as the format, then click Export.

Tip: Work at 512 × 512 or 1024 × 1024 pixels for the best balance of quality and performance. If GIMP opens the file at a different size, do not resize it — paint within the existing canvas.

1.4 Copy the edited texture back into the container

```
podman cp car1-trb1.rgb
<container_name>:/usr/local/share/games/torcs/drivers/src_server/0/car1-
trb1.rgb
```

Restart TORCS. Your car will now appear with the new livery on track.

✓ The texture file is back in the container
✓ TORCS has been restarted
✓ Your car appears in the new colours on the track selection screen

2. Adding Multiple Livery Variants

TORCS supports multiple liveries for the same car model — useful if your team has several drivers, or if you want a different look for qualifying versus the race.

2.1 Create the additional texture files

Name the files using the same base name with a number suffix:

```
car1-trb1.rgb      ← driver slot 1 (default)
car1-trb1-2.rgb   ← driver slot 2
car1-trb1-3.rgb   ← driver slot 3
```

Each file is an independent texture — design them differently if you wish.

2.2 Point a driver to a specific livery

Open the driver XML file for your robot. This is found at:

```
/usr/local/share/games/torcs/drivers/<robotname>/<robotname>.xml
```

Inside the file, find the section for each driver entry. Add or update the tex name attribute to reference your chosen skin:

```
<attstr name="tex name" val="car1-trb1-2"/>
```

Save the file, copy it back into the container with podman cp, and restart TORCS.

3. Customising Track Banners

Banners are the large advertising boards and trackside signs you see around the circuit. Each one is a flat texture mapped onto a simple rectangular object in the 3D scene. Replacing the texture file changes what is displayed.

3.1 Find the banner textures

Track files are stored at:

```
/usr/local/share/games/torcs/tracks/<category>/<trackname>/
```

For example, for the road track aalborg:

```
/usr/local/share/games/torcs/tracks/road/aalborg/
```

Look for files named banner.rgb, advert1.rgb, or similar. Copy one out to inspect it before editing — banners are typically wide and short (e.g. 512 × 128 pixels).

3.2 Design your banner in GIMP

5. Open GIMP and then open the .rgb banner file.
6. Replace the artwork with your own design — team name, university logo, or a motivational slogan.
7. Export As the same filename in SGI Image (.rgb) format.
8. Copy the file back into the container using podman cp, placing it in the same track folder.

> 💡 **Tip:** Banner textures are often very wide and short. Keep your design horizontal and avoid fine text smaller than about 30 pixels tall — it will be hard to read at racing speed.

3.3 Verify the track XML (optional)

If your banner does not appear after restarting, open the track's XML file (e.g. aalborg.xml) and search for the banner filename. It will look something like this:

```
<attstr name="texture" val="banner.rgb"/>
```

Make sure the val attribute matches your file's actual name, including the extension. If you renamed the file, update this entry to match.

4. Quick Troubleshooting

Problem	Fix
Car still shows old colours	Restart TORCS fully — textures are cached at launch and will not reload during a session.
Texture appears black or corrupt	Check that the image dimensions are powers of two (256, 512, 1024). Resize the canvas in GIMP if needed.
GIMP cannot open the .rgb file	Ensure you have GIMP 2.10 or later. Go to File > Open and select All Files to see .rgb files in the browser.
Banner does not update on track	Confirm the podman cp command completed without error, then restart TORCS and select the same track again.
Second livery variant not shown	Check the driver XML file and confirm the tex name attribute matches the filename exactly (without the .rgb extension).

Need more help?
Post a screenshot of your car or banner in the competition chat channel — the teaching team and your fellow students will be happy to assist. Include the output of any error messages shown in the TORCS console window.